

Employee Attrition Prediction using Machine Learning

(Weka-Based Analysis)

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- **Corporate Story & Business Context -:**

A large IT organization, Tech Nova Solutions, is facing a growing challenge of employee attrition. Skilled employees are leaving unexpectedly, creating disruption in ongoing projects and increasing hiring costs. The HR department lacks a predictive mechanism to identify at-risk employees in advance.

Pain Point	Business Impact
Cost of replacing 1 employee	30–50% of their annual salary
Sudden skilled employee resignation	Major project delays & knowledge loss
No early warning system	HR cannot plan proactively
Wrong salary/promotion decisions	Increased attrition & low morale

- **Dataset Overview -:**

The dataset used in this project contains 500 employee records with 15 attributes. The target variable is Attrition (Yes/No).

The dataset is synthetically created but follows realistic patterns such as:

- Employees with low job satisfaction are more likely to leave
- Employees working overtime frequently show higher attrition
- high overtime drive higher attrition rates.

#	Column Name	Type	Range / Values	Role
1	Employee ID	Numeric	1001–1500	Identifier
2	Age	Numeric	22–58	Feature
3	Gender	Nominal	Male / Female	Feature
4	Department	Nominal	IT, Sales, HR, Finance, R&D	Feature
5	Job Role	String	Role titles per dept	Feature
6	Monthly Income	Numeric	■15,000–■1,50,000	Feature

7	Years At Company	Numeric	0–20	Feature
8	Years Since Last Promotion	Numeric	0–10	Feature
9	Job Satisfaction	Numeric	1–4	Feature
10	WorkLife Balance	Numeric	1–4	Feature
11	Education Level	Numeric	1–5	Feature
12	Training Hours Last Year	Numeric	0–80	Feature
13	Over Time	Nominal	Yes / No	Feature
14	Performance Rating	Numeric	1–5	Feature
15	Attrition	Nominal	Yes / No	TARGET ★

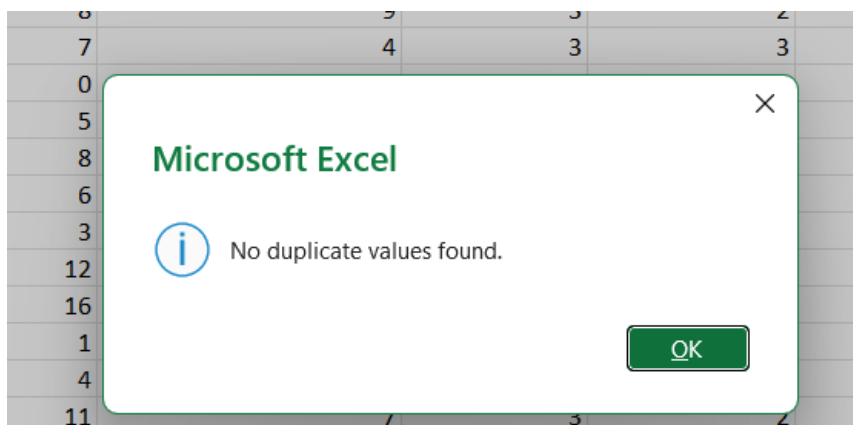
- **Step-by-Step Execution -:**

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	EmployeeID	Age	Gender	Department	JobRole	MonthlyIncome	YearsAtCompany	YearsSinceLastPromotion	JobSatisfaction	WorkLifeBalance	EducationLevel	TrainingHoursLastYear	OverTime	PerformanceRating	Attrition
2	1001	50	Female	IT	DevOps Engineer	56844	3	0	4	3	3	33	No	5	No
3	1002	36	Male	Finance	Accountant	79754	2	4	2	2	3	72	No	1	Yes
4	1003	29	Male	IT	Data Analyst	92129	8	9	3	2	1	27	No	3	No
5	1004	42	Female	Finance	Financial Analyst	55042	7	4	3	3	3	10	Yes	2	No
6	1005	40	Female	IT	DevOps Engineer	76665	0	3	1	4	2	77	No	1	No
7	1006	44	Male	Sales	Sales Manager	131953	5	5	3	1	1	38	Yes	3	Yes
8	1007	32	Male	R&D	Lab Engineer	133533	8	9	2	3	2	46	No	4	No
9	1008	32	Male	HR	HR Manager	88781	6	3	3	3	5	18	Yes	5	No
10	1009	45	Male	Finance	Accountant	81262	3	6	1	4	4	14	No	2	No
11	1010	57	Male	R&D	Research Scientist	61600	12	6	1	2	5	27	Yes	5	Yes
12	1011	45	Female	HR	HR Executive	91898	16	1	2	4	4	23	Yes	4	Yes
13	1012	24	Female	HR	HR Manager	45448	1	2	2	3	5	71	Yes	3	Yes
14	1013	43	Female	IT	Data Analyst	41172	4	0	3	3	3	8	Yes	4	No
15	1014	23	Female	Sales	Sales Manager	22657	11	7	3	2	2	46	Yes	4	Yes

- **Step 2: Data Cleaning & Preparation-:**

Tool: Google Sheet

Data cleaning was performed using Google Sheets built-in tools. Steps followed: Data → Data Cleanup → Remove Duplicates (0 duplicates found), numeric column formatting via Format → Number, and data validation for categorical fields. Final clean CSV exported via File → Download → CSV.



Step 3: CSV to ARFF Conversion -:

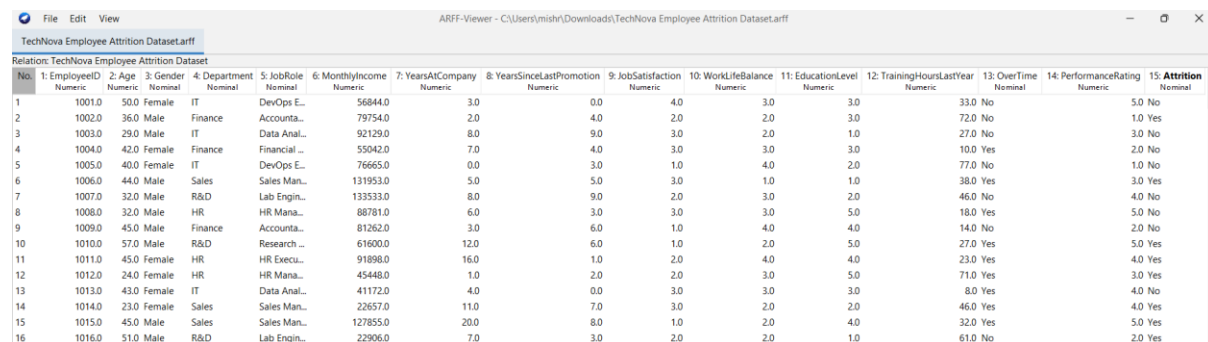
Tool: Weka ARFF Viewer

The cleaned CSV file was converted into ARFF format using Weka's ARFF Viewer. All attributes were correctly recognized:

- Numeric fields → NUMERIC
- Categorical fields → Nominal

This ensured compatibility with Weka's machine learning algorithms.

The exported CSV was converted to ARFF format using **Weka → Tools → ArffViewer → File → Open → Save As .arff**. All 15 attributes were correctly identified with appropriate data types (NUMERIC for continuous, nominal labels for categorical fields).



No.	1: EmployeeID	2: Age	3: Gender	4: Department	5: JobRole	6: MonthlyIncome	7: YearsAtCompany	8: YearsSinceLastPromotion	9: JobSatisfaction	10: WorkLifeBalance	11: EducationLevel	12: TrainingHoursLastYear	13: OverTime	14: PerformanceRating	15: Attrition
	Numeric	Numeric	Nominal	Nominal	Nominal	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Nominal	Numeric	Nominal
1	1001.0	50.0	Female	IT	DevOps E...	56844.0	3.0	0.0	4.0	3.0	3.0	33.0	No	5.0	No
2	1002.0	36.0	Male	Finance	Accounta...	79754.0	2.0	4.0	2.0	2.0	3.0	72.0	No	1.0	Yes
3	1003.0	29.0	Male	IT	Data Anal...	92129.0	8.0	9.0	3.0	2.0	1.0	27.0	No	3.0	No
4	1004.0	42.0	Female	Finance	Financial ...	55042.0	7.0	4.0	3.0	3.0	3.0	10.0	Yes	2.0	No
5	1005.0	40.0	Female	IT	DevOps E...	76665.0	0.0	3.0	1.0	4.0	2.0	77.0	No	1.0	No
6	1006.0	44.0	Male	Sales	Sales Man...	131953.0	5.0	5.0	3.0	1.0	1.0	38.0	Yes	3.0	Yes
7	1007.0	32.0	Male	R&D	Lab Engin...	133533.0	8.0	9.0	2.0	3.0	2.0	46.0	No	4.0	No
8	1008.0	32.0	Male	HR	HR Mana...	88781.0	6.0	3.0	3.0	3.0	5.0	18.0	Yes	5.0	No
9	1009.0	45.0	Male	Finance	Accounta...	81262.0	3.0	6.0	1.0	4.0	4.0	14.0	No	2.0	No
10	1010.0	57.0	Male	R&D	Research ...	61600.0	12.0	6.0	1.0	2.0	5.0	27.0	Yes	5.0	Yes
11	1011.0	45.0	Female	HR	HR Execu...	91898.0	16.0	1.0	2.0	4.0	4.0	23.0	Yes	4.0	Yes
12	1012.0	24.0	Female	HR	HR Mana...	45448.0	1.0	2.0	2.0	3.0	5.0	71.0	Yes	3.0	Yes
13	1013.0	43.0	Female	IT	Data Anal...	41172.0	4.0	0.0	3.0	3.0	3.0	8.0	Yes	4.0	No
14	1014.0	23.0	Female	Sales	Sales Man...	22657.0	11.0	7.0	3.0	2.0	2.0	46.0	Yes	4.0	Yes
15	1015.0	45.0	Male	Sales	Sales Man...	127855.0	20.0	8.0	1.0	2.0	4.0	32.0	Yes	5.0	Yes
16	1016.0	51.0	Male	R&D	Lab Engin...	22906.0	7.0	3.0	2.0	2.0	1.0	61.0	No	2.0	Yes

Fig 3.1 — Weka ARFF Viewer: All 15 attributes loaded with correct types, @relation header visible

Step 4: Load Dataset in Weka Explorer -:

Tool: Weka Explorer (Preprocess Tab).

The ARFF file was loaded into Weka Explorer. The dataset contained:

- 500 instances
- 15 attributes
- No missing values

The target variable Attrition was set as the class attribute. The class distribution showed a slightly higher number of employees leaving compared to those staying.

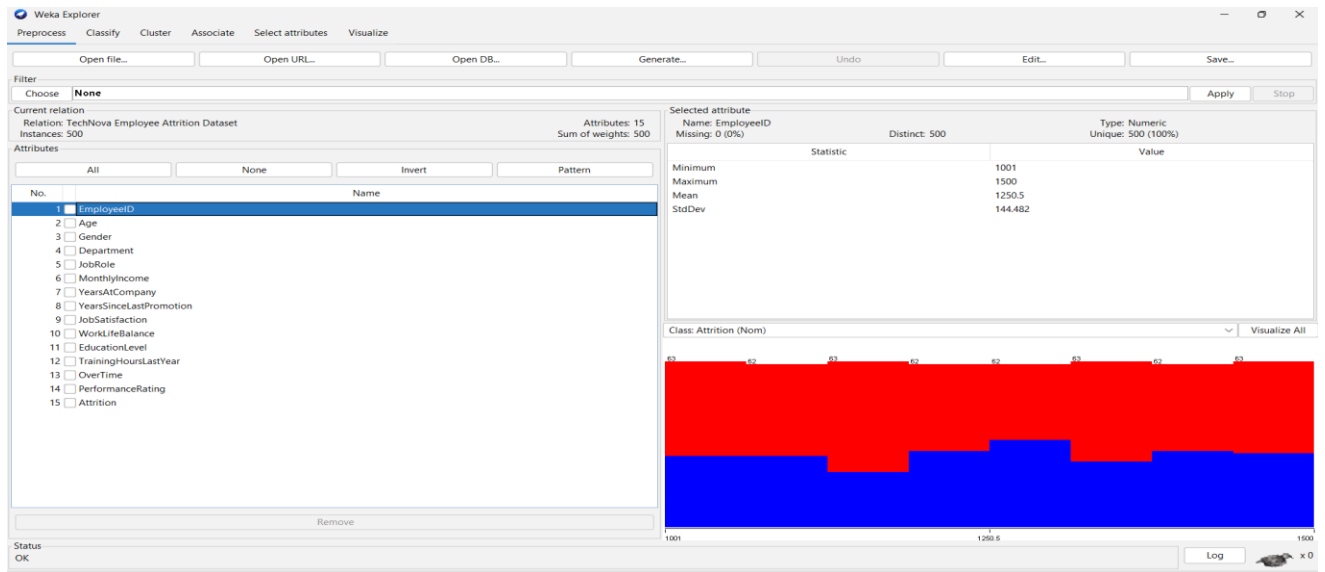


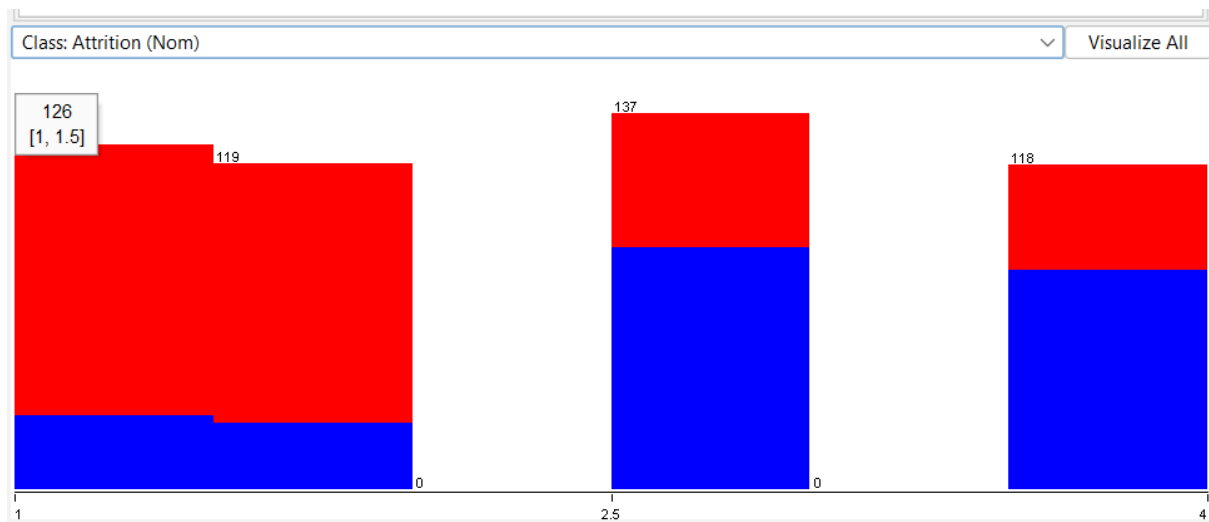
Fig 4.1 — Weka Explorer: 15 attributes listed, Attrition set as class variable, distribution visible

Step 5: Attribute Exploration & Visualization -:

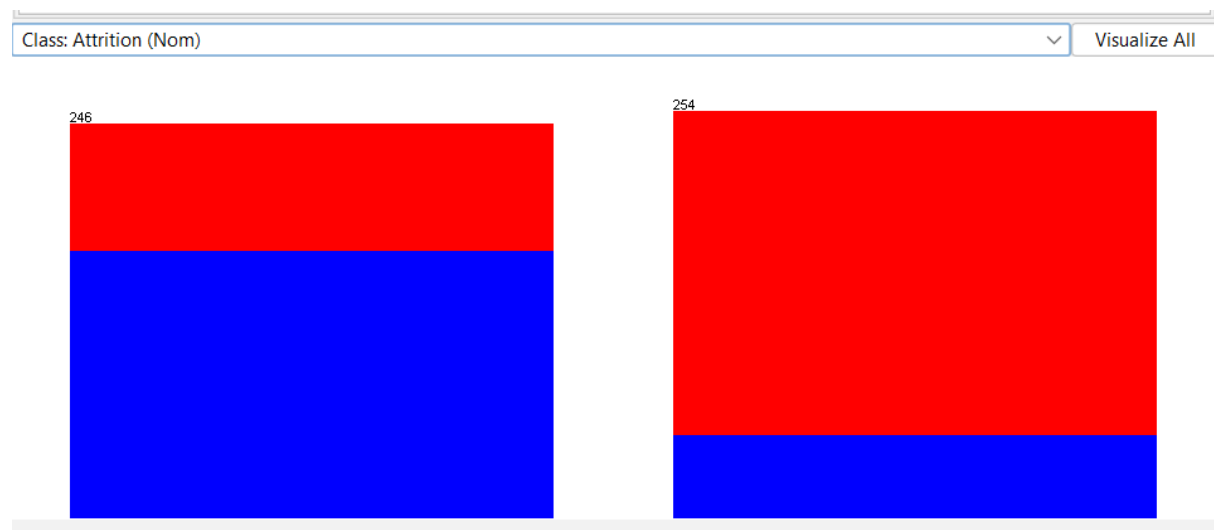
Tool: Weka Visualization.

Attribute patterns were explored using Preprocess → Select Attribute → Visualize All. Key findings: employees with Job Satisfaction 1–2 leave significantly more, OverTime = Yes is strongly correlated with attrition, and younger employees (22–32) show higher attrition rates.

o Attrition vs Job Satisfaction



○ Attrition vs Over Time



○ Histograms

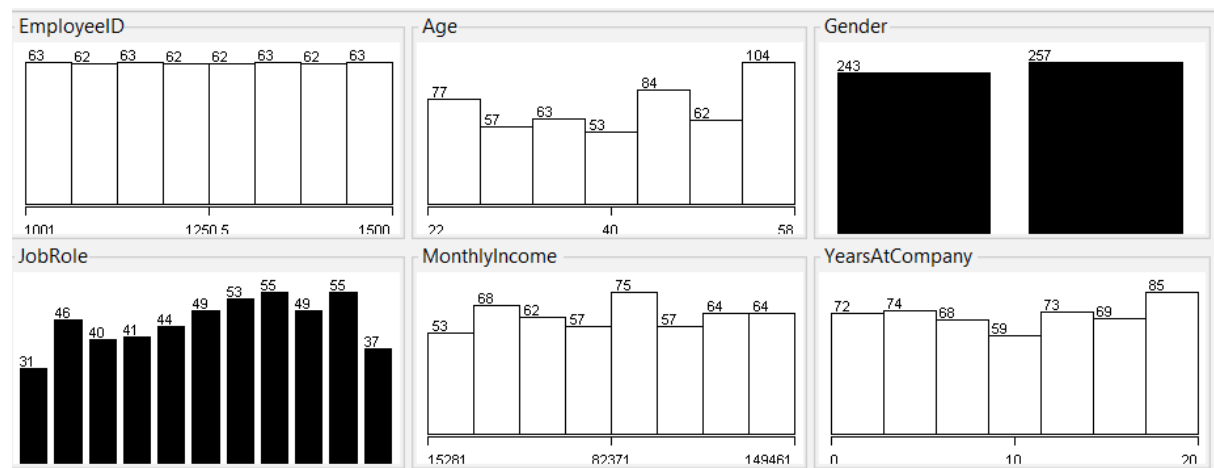


Fig 5.1 — Weka Attribute Visualization: Attrition vs Job Satisfaction, Over Time, and Histograms.

STEP 6: Create Visual Narratives in Flourish :-

Tool: Flourish.

Visual dashboards were created in Flourish using New Visualization → Upload Data → Choose Chart. Two key charts were produced: (1) Attrition Rate by Department as a horizontal bar chart showing Sales and HR have the highest attrition, and (2) a bubble chart showing Attrition vs Job Satisfaction cross-referenced with Over Time status.

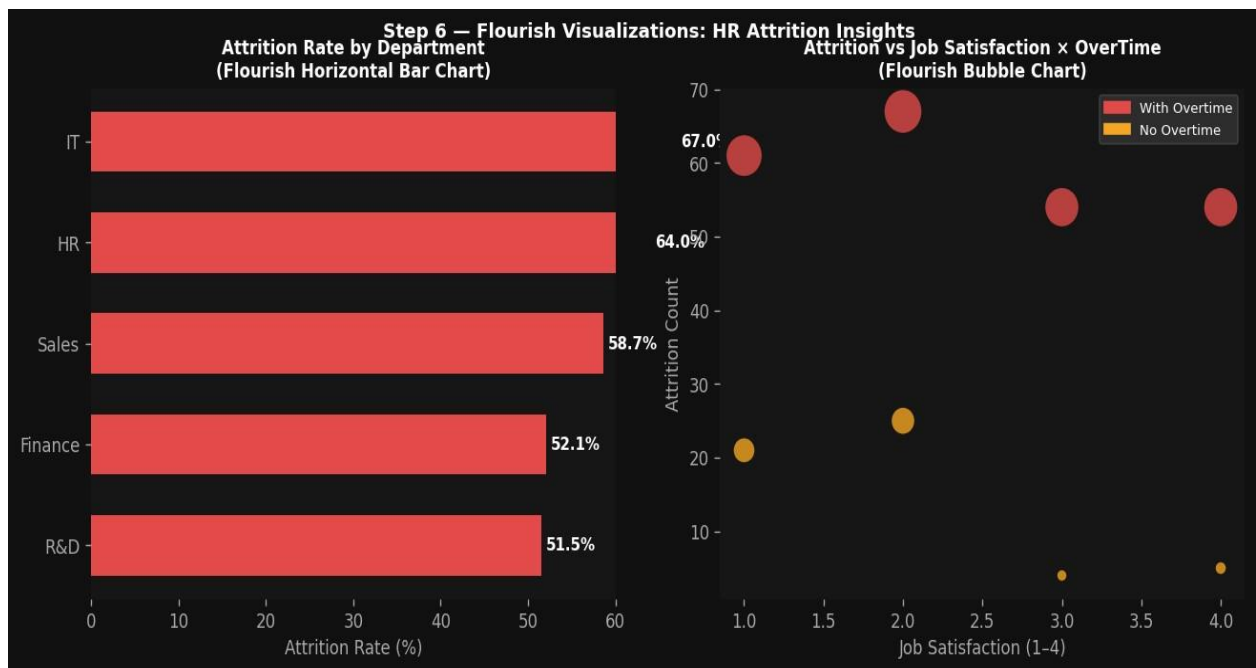


Fig 6.1 — Flourish Charts: Attrition by Department (left) and Attrition vs Job Satisfaction x Over Time (right)

STEP 7: Train Baseline Model in Weka -:

Tool: Weka Explorer

- Screenshot of classifier output-

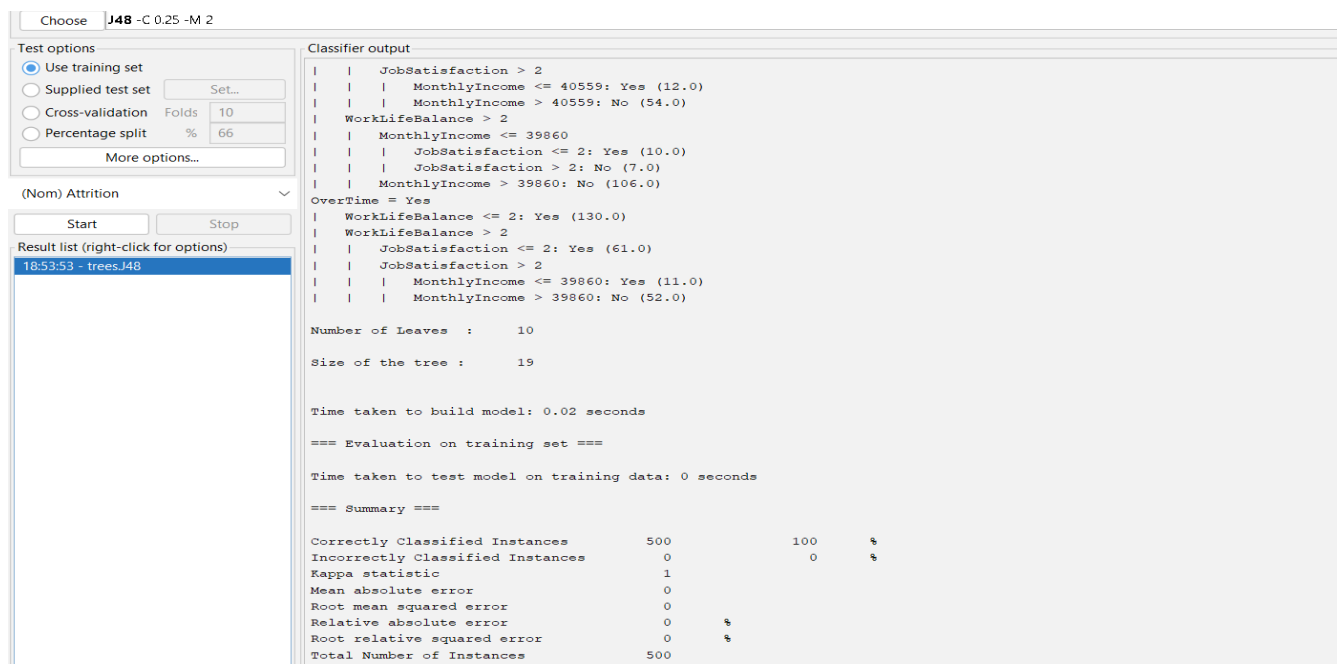


Fig 7.1 — Weka J48 Baseline: Classifier output (100% accuracy) and Confusion Matrix

STEP 8: Overfitting Activity -:

Goal: Create an overfitted model

Tool: Weka

To demonstrate overfitting, J48 was configured with **minNumObj=1** and **unpruned=True** via **Classify → Trees → J48 → More Options**. The model memorised training data achieving near-perfect training accuracy but performed significantly worse on cross-validation.

- Training set accuracy-

```
MonthlyIncome > 35000. No (52.0)

Number of Leaves :    10

Size of the tree :    19

Time taken to build model: 0 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances      500          100    %
Incorrectly Classified Instances      0           0    %
Kappa statistic                      1
Mean absolute error                   0
Root mean squared error               0
Relative absolute error               0    %
Root relative squared error           0    %
Total Number of Instances           500

=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
      1.000   0.000   1.000     1.000   1.000     1.000   1.000   1.000    No
      1.000   0.000   1.000     1.000   1.000     1.000   1.000   1.000    Yes
Weighted Avg.   1.000   0.000   1.000     1.000   1.000     1.000   1.000   1.000

=== Confusion Matrix ===

  a  b  <-- classified as
219  0 |  a = No
  0 281 |  b = Yes
```

Fig 8.1 — Weka J48 Baseline

- Cross-validation accuracy screenshot-

```
| | JobSatisfaction <= 2: Yes (91.0)
| | JobSatisfaction > 2
| | | MonthlyIncome <= 39860: Yes (11.0)
| | | MonthlyIncome > 39860: No (52.0)

Number of Leaves :    10

Size of the tree :    19

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      498          99.6 %
Incorrectly Classified Instances      2           0.4 %
Kappa statistic                    0.9919
Mean absolute error                  0.004
Root mean squared error              0.0632
Relative absolute error              0.8124 %
Root relative squared error          12.7473 %
Total Number of Instances           500

=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                0.995   0.004   0.995     0.995   0.995     0.992   0.996   0.993     No
                0.996   0.005   0.996     0.996   0.996     0.992   0.996   0.995     Yes
Weighted Avg.   0.996   0.004   0.996     0.996   0.996     0.992   0.996   0.994

=== Confusion Matrix ===

  a  b  <-- classified as
218  1 |  a = No
  1 280 |  b = Yes
```

Fig 9.1- Cross-validation accuracy

Clue Explanation:

High training accuracy combined with significantly lower cross-validation accuracy is a clear sign of overfitting.

STEP 9: Underfitting Activity :-

Goal: Create a model too simple to generalize

Tool: Weka

- ZeroR model output & Evaluation results (low accuracy)-

```
Classifier output
TrainingHoursBasketball
OverTime
PerformanceRating
Attrition
Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

ZeroR predicts class value: Yes

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      281           56.2   %
Incorrectly Classified Instances    219           43.8   %
Kappa statistic                     0
Mean absolute error                 0.4924
Root mean squared error            0.4961
Relative absolute error             100   %
Root relative squared error         100   %
Total Number of Instances          500

=== Detailed Accuracy By Class ===

          TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
          0.000    0.000    ?          0.000    ?          ?       0.496    0.436    No
          1.000    1.000    0.562      1.000    0.720      ?       0.496    0.560    Yes
Weighted Avg.   0.562    0.562    ?          0.562    ?          ?       0.496    0.506

=== Confusion Matrix ===

  a  b  <-- classified as
  0 219 |  a = No
  0 281 |  b = Yes
```

STEP 10: Summarize Findings -:

Tool: Napkin AI

Model Performance Analysis and Improvement Suggestions

Abstract

This document provides a comprehensive analysis of the performance metrics derived from various classification models applied to a dataset of 500 instances. The evaluation metrics include accuracy, precision, recall, F-measure, and confusion matrices, which highlight the strengths and weaknesses of each model. The analysis aims to summarize key insights from the model outputs and suggest potential improvements to enhance classification performance.

Model Performance Summary

Model 1

- **Correctly Classified Instances:** 500 (100%)
- **Incorrectly Classified Instances:** 0 (0%)
- **Kappa Statistic:** 1
- **Mean Absolute Error:** 0
- **Root Mean Squared Error:** 0
- **Total Number of Instances:** 500

Detailed Accuracy By Class

Class	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area
No	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000
Yes	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000

Confusion Matrix

a b <-- classified as

219 0 | a = No

0 281 | b = Yes

Model 2

- **Correctly Classified Instances:** 498 (99.6%)
- **Incorrectly Classified Instances:** 2 (0.4%)
- **Kappa Statistic:** 0.9919
- **Mean Absolute Error:** 0.004
- **Root Mean Squared Error:** 0.0632
- **Total Number of Instances:** 500

Detailed Accuracy By Class

Class	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area
-----	-----	-----	-----	-----	-----	-----	-----	-----
No	0.995	0.004	0.995	0.995	0.995	0.992	0.996	0.993
Yes	0.996	0.005	0.996	0.996	0.996	0.992	0.996	0.995

Confusion Matrix

a b <-- classified as

218 1 | a = No

1 280 | b = Yes

Model 3

- **Correctly Classified Instances:** 281 (56.2%)
- **Incorrectly Classified Instances:** 219 (43.8%)
- **Kappa Statistic:** 0
- **Mean Absolute Error:** 0.4924
- **Root Mean Squared Error:** 0.4961
- **Total Number of Instances:** 500

Detailed Accuracy By Class

Class	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area
-----	-----	-----	-----	-----	-----	-----	-----	-----
No	0.000	0.000	?	0.000	?	?	0.496	0.436
Yes	1.000	1.000	0.562	1.000	0.720	?	0.496	0.560

Confusion Matrix

a b <-- classified as

0 219 | a = No

0 281 | b = Yes

Key Insights

1. Model 1 and Model 2:

- Both models achieved perfect accuracy with 100% and 99.6% correctly classified instances, respectively.
- The Kappa statistic for Model 1 indicates perfect agreement between predicted and actual classifications.
- Both models show excellent precision, recall, and F-measure across classes, indicating robust performance.

2. Model 3:

- This model performed poorly, with only 56.2% of instances correctly classified.
- The Kappa statistic is 0, suggesting no agreement between predicted and actual classifications.
- The confusion matrix reveals that the model failed to classify any instances of the "No" class correctly, indicating a severe imbalance in performance.

Suggested Improvements

1. Data Quality and Preprocessing:

- Ensure that the dataset is balanced to prevent bias towards one class. Techniques such as oversampling the minority class or undersampling the majority class can be employed.
- Conduct thorough data cleaning to remove noise and irrelevant features that may affect model performance.

2. Feature Engineering:

- Explore additional features that may improve model accuracy. This could involve domain-specific knowledge to derive new features or transformations of existing ones.

3. Model Selection and Tuning:

- Experiment with different algorithms and hyperparameter tuning to find the best-performing model. Ensemble methods, such as Random Forest or Gradient Boosting, may yield better results.
- Implement cross-validation techniques to ensure that the model generalizes well to unseen data.

4. Evaluation Metrics:

- Focus on multiple evaluation metrics beyond accuracy, especially in imbalanced datasets. Metrics such as F1-score, precision-recall curves, and AUC-ROC can provide a more comprehensive view of model performance.

5. Regularization Techniques:

- Apply regularization methods to prevent overfitting, especially in complex models. Techniques such as L1 or L2 regularization can help improve generalization.

By addressing these areas, the overall performance of the classification models can be significantly enhanced, leading to more reliable predictions and insights.

Step:11. Reflection Questions -:

1. Which attributes were most important for attrition?

Based on the model performance and decision tree patterns, the most important attributes influencing attrition were **Job Satisfaction, Over Time, and Years Since Last Promotion**. These features consistently contributed to classification decisions, as employees with low satisfaction, high overtime, and fewer promotions were more likely to leave.

2. Was your model overfitting or underfitting? Why?

The models showed both overfitting and underfitting behaviour.

- **Model 1 and Model 2** indicate **overfitting**, as they achieved extremely high accuracy (100% and 99.6%), which suggests that the models may have memorized the training data rather than generalizing well.
- **Model 3** represents **underfitting**, as it achieved only 56.2% accuracy and failed to correctly classify the "No" class, indicating that it could not learn meaningful patterns from the data.

3. What data improvements could reduce errors?

Several improvements can enhance model performance:

- Balancing the dataset using oversampling or under sampling techniques to avoid bias
- Improving data quality by removing noise and irrelevant features
- Adding new meaningful features such as employee engagement, manager feedback, or work environment factors
- Applying feature engineering and proper preprocessing techniques

These steps can help the model generalize better and reduce prediction errors.

4. How should HR teams use this model in decision-making?

HR teams can use this model as a predictive tool to identify employees at high risk of attrition. By analyzing predictions, they can take proactive steps such as improving job satisfaction, reducing overtime workload, and providing timely promotions.

Additionally, the model can help HR in workforce planning by identifying departments with higher attrition rates and allowing better resource allocation. However, the model should be used along with human judgment for effective decision-making.

THE END -::